



XVI Simpósio Brasileiro de Melhoramento Animal

Piracicaba, SP – 04 e 05 de agosto de 2025

Discovering structural variations in the Nelore genome using long-read sequencing

Thales de Lima Silva^{*1}, Haroldo Henrique de Rezende Neves², Alexander S. Leonard³, Gerardo A. Fernandes Júnior⁴, Júlio Cesar de Carvalho Balieiro¹, Roberto Carneiro⁵, Hubert Pausch³ and Ricardo Vieira Ventura¹

¹Universidade de São Paulo, Pirassununga/SP, Brasil; ²GenSys Consultores Associados, Porto Alegre/RS, Brasil; ³Eidgenössische Technische Hochschule, Zürich, Switzerland; ⁴Universidade Estadual Vale do Acaraú, Sobral/CE, Brazil; ⁵Commonwealth Scientific and Industrial Research Organisation, Hobart/TAS, Australia.

*Corresponding author - tl.silva@usp.br

The tropical climate imposes selective pressures on cattle, driving genomic adaptations that remain incompletely understood. Breeds such as Nelore cattle (*Bos indicus*), known for their short, slick coats, heat tolerance, and resilience, exemplify these adaptations. Its characteristics make this breed a valuable model for investigating the genetic basis of thermotolerance, a complex and heritable trait that can be improved through selection. However, most current studies rely on SNP arrays developed for taurine breeds, which may not fully capture the genetic diversity of indicine cattle. Structural variants (SV) are increasingly recognized as important contributors to complex trait variation. However, due to the limited resolution of short read sequencing technologies, their prevalence and functional impact in bovine genomes remain largely uncharacterized. Long-read sequencing allows for more accurate detection of SV, although their application in bovine genomics is still emerging. In this study, we used Oxford Nanopore Technologies long-read data from 20 Nelore bulls (COWADAPT project) to identify and characterize SV using tools such as Minimap2 for alignment and Sniffles2 for SV discovery and genotyping. To contextualize our findings, we compared the detected SVs to those called in a publicly available dataset from a Nelore × Brown Swiss individual (ERR7091271) deposited at the European Nucleotide Archive (ENA). Analyses were restricted to autosomal chromosomes. SV annotation was performed in R using VariantAnnotation package to process VCF files, with gene-level annotations derived from org.Bt.eg.db and genomic coordinates mapped to the TxDb.Btaurus.UCSC.bosTau9.refGene assembly. Visualization leveraged BSgenome.Btaurus.UCSC.bosTau9 for sequence context analysis. SV statistics were derived from the SVLEN and SVTYPE fields in the VCF files. Overall, the number of SVs detected in the COWADAPT was comparable to the ERR7091271 sample. Insertions and deletions accounted for over 99% of total SV, with mean counts ($\pm$ standard deviation) of $26k \pm 0.5k$ and $25k \pm 0.6k$, respectively. Among these, we identified 179 SV (104 deletions and 75 insertions) shared across all samples that overlapped protein-coding regions, providing a focused set of candidate genes for investigating adaptive traits in Nelore cattle. Functional annotation highlighted recurrent SV in annotated genes, including an 84 bp deletion in *PRDM9* and insertions in *SLC66A1* (60 bp), *RNF32* (70 bp), *ACER2* (102 bp), and *PHGR1* (72 bp). These findings underscore the value of long-read sequencing in uncovering large, potentially functional genomic variants and provide a foundation for future functional validation and selection strategies in indicine cattle. Future studies with larger sample sizes are needed to validate these findings and establish if these genes are targets for improving adaptive traits in Nelore cattle.

Key-words: COWADAPT, structural variant calling, tropical adaptation.

Acknowledgements: This work was financially supported by São Paulo Research Foundation (FAPESP - Grants nº 2021/03101-9, 2023/06637-2 and 2024/18544-1).